# EE5203 Microprocessors — Syllabus

2026-2027 · Istanbul Kültür University · Electrical & Electronics Engineering

## What this course is

EE5203 develops embedded-system competence through a C-first STM32 pathway. You first build reliable Embedded C foundations, then learn enough register-level programming to understand the machine, and finally use the STM32 HAL to implement increasingly complex peripherals and an integrated project of your own.

Four ideas shape every week:

1. **Embedded C comes first.** Four full lectures build the C you need; everything later assumes it.
2. **Register literacy, HAL fluency.** You implement GPIO directly on registers once, so that when you use HAL afterwards you know what it does.
3. **Labs fade their scaffolding.** Early labs walk you through the workflow; later labs hand you a specification and expect a working, explained result.
4. **The project starts early.** You see the destination in Lecture 1, form teams mid-term, and demonstrate working subsystems as milestones — not one big demo at the end.

## Learning outcomes

By the end of the course, a successful student can:

1. Write, compile, debug, and test Embedded C programs.
2. Explain the Cortex-M memory, register, and exception model.
3. Configure GPIO, timers, PWM, ADC, UART, and a synchronous serial peripheral using an appropriate combination of CMSIS register access and STM32 HAL.
4. Design interrupt-driven, non-blocking embedded applications.
5. Interpret datasheets, reference manuals, timing diagrams, compiler messages, debugger state, and peripheral registers as engineering evidence.
6. Integrate, test, document, demonstrate, and individually explain an STM32-based embedded system.

## Tools and hardware

| Item | What we use |
|---|---|
| Board | ST Nucleo-F401RE (provided in the laboratory) |
| Configuration | STM32CubeMX (pin configuration and project generation) |
| Development | VS Code with the course STM32 profile (edit, build, flash, debug) |
| Where things run | Laboratory PCs are fully set up; personal installation instructions are on the L01 lab reference page |

## Weekly rhythm

Each lecture package on this site has the same parts, used in the same order:

1. **Prepare** — a short study task and the pre-lab quiz (taken in CATS, two attempts, highest score counts).
2. **Learn** — the theory session and its study sheet.
3. **Practise** — a 30-45 minute worksheet; worked solutions are released in CATS after the corresponding session.
4. **Lab** — the lab guide, with an individual checkoff and a short oral check. Every lab follows the same cycle: warm-up, foundation checkpoint, independent core task, individual modification, oral check.
5. **Review** — a mastery checklist and exam-style questions inside the study sheet.

Graded pre-lab quizzes and homework are **only** in CATS — they are not published on this site. Announcements, submissions, and the weekly calendar schedule also live in CATS.

## Course sequence

Twelve teaching weeks: eleven lecture weeks and a midterm week. Week 8 is the midterm and introduces no new material. UART and I2C are taught together in week 11 (SPI is summarised there and continues in the project track), and the integration clinic runs in the laboratory hours of weeks 11 and 12.

| ID | Week | Topic | Central performance | Project step |
|---|---|---|---|---|
| L01 | 1 | Embedded systems, C toolchain, and debugging | Build, flash, debug, and modify Blinky | See example projects |
| L02 | 2 | Values, decisions, loops, and arrays | Process a sensor-data array in one C file | Identify project inputs |
| L03 | 3 | Functions, arrays, strings, and modular C | Decompose the sensor program into functions and files | Identify processing functions |
| L04 | 4 | Pointers, structures, bits, registers, and GPIO | Control PA5 through CMSIS registers | Individual idea canvas |
| L05 | 5 | GPIO and hardware abstraction | Rebuild GPIO with HAL and compare the two interfaces | Teams form; project launch |
| L06 | 6 | Interrupts and the NVIC | Implement and explain an event-driven application | Proposal and block diagram |
| L07 | 7 | Timers and non-blocking time | Generate periodic events without blocking delays | — |
| | **8** | **Midterm examination** | | |

| ID | Week | Topic | Central performance | Project step |
|---|---|---|---|---|
| L08 | 9 | PWM, output compare, and input capture | Generate timed signals — and measure them | Design review; first hardware milestone |
| L09 | 10 | ADC and sensor acquisition | Acquire, scale, and validate analog data | Data-acquisition milestone |
| L10 | 11 | Serial communication: UART and I2C (SPI summary) | Stream data to a PC, accept commands, read one I2C device | Communication milestone |
| L11 | 11 | I2C and SPI — taught with L10 | Interface a synchronous serial device | Sensor/display integration |
| L12 | 11–12 (lab) | Integration clinic in laboratory hours | Integrate peripherals and diagnose faults | Working prototype |
| L13 | 12 | Demonstration and defense | Demonstrate, test, and individually explain the system | Final assessment |

Week numbers are the planned sequence; the authoritative weekly calendar, including any holiday shifts, is always the one announced in CATS.

## Assessment

Assessment combines a midterm (including an individual toolchain practical), laboratory performance (individual checkoffs plus pre-lab quizzes and homework), the term project (proposal, milestones, integration, demo, report, and oral defense), and a final examination. Official weights follow the ECTS course record; any clarifications are announced in CATS.

Laboratory credit reflects **directly observed performance**: checkoffs are individual, and every lab ends with a short oral check where you explain your own work.

## Project timeline

| Point | Week | Milestone |
|---|---|---|
| L04 | 4 | One-page individual idea canvas |
| L05 | 5 | Teams form; preliminary topic |
| L06 | 6 | Proposal: problem, block diagram, components, minimum viable system |
| L08 | 9 | Design review (pins, peripherals, data flow, risks) and first hardware milestone |
| L09 | 10 | Acquired and validated sensor data |
| L10 | 11 | Communication milestone — UART logging or command interface |
| L11 | 11 | Integration milestone — input, processing, output, communication connected |
| L12 | 11–12 (lab) | Readiness review: working prototype and test evidence |
| L13 | 12 | Demonstration, report, presentation, and individual oral defense |

There is no milestone in week 8. That week is yours for the midterm.